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Chapter 2 Random variable and their probability distributions

Section 2.2 Discrete random variable and its distribution law

- Distribution function
- Distribution law of discrete random variable
- Common distribution function of discrete random variable
- Examples

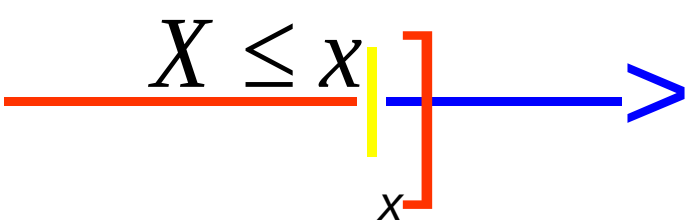
Distribution function

Definition 2.1.2: Let $\boldsymbol{X}$ be a $r.v$, then

$$\boldsymbol{F(x) = P(X \le x) \quad (-\infty < x < +\infty)}$$

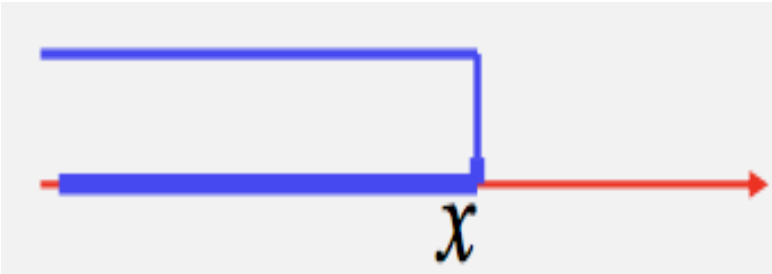
is called the distribution function of $\boldsymbol{X}$, referred as $\boldsymbol{X \sim F(x)}$ or $\boldsymbol{F_X(x)}$.

If $\boldsymbol{X}$ is regarded as a random points in the x axis, then distribution function F(x) denotes the probability that X falls in $(-\infty, x]$.



Distribution function

Consider X as the coordinate of a random point on the number axis:



- $P\{X \leq 1\}$ corresponds to 1, and $P\{X \leq 2\}$ corresponds to 2;
- **$P\{X \leq x\}$ corresponds to x, so $P\{X \leq x\}$ is a function of x ;**

➢ After introducing the distribution function $F(x)$, the probability of an event can be represented by the function value of $F(x)$

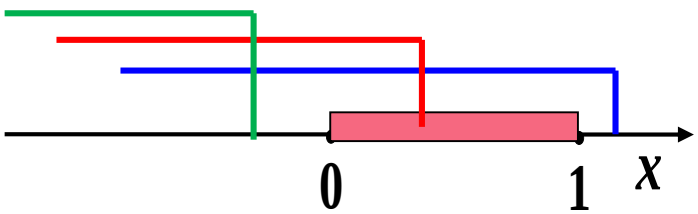
Distribution law of discrete random variable

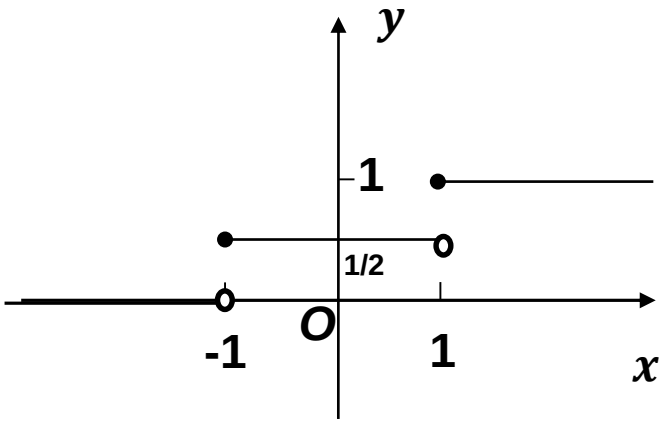
Example 2.2.1: Tossing a fair coin, Let

$$X = \begin{cases} 1, & frontage\ on\ the\ top \\ 0, & reverse\ side\ on\ the\ top \end{cases}$$

Calculate the distribution function of X

Answer:





$$\{\boldsymbol{X} \le \boldsymbol{x}\} = \begin{cases} \emptyset, & x < 0, \\ \{0\}, & 0 \le x < 1, \\ \{0,1\}, & x \ge 1, \end{cases} \qquad \boldsymbol{F}(\boldsymbol{x}) = \begin{cases} 0, & x < 0, \\ \frac{1}{2}, & 0 \le x < 1, \\ 1, & x \ge 1, \end{cases}$$

Distribution function

The properties of distribution function:

(1) $0 \le F(x) \le 1, (-\infty < x < +\infty)$;

$$F(-\infty)=\lim_{x\to-\infty} F(x) = 0, F(\infty)=\lim_{x\to\infty} F(x) = 1$$

(2) $F(x_1) \le F(x_2), x1 < x2$

(3) $\lim_{x\to x_0^+} F(x) = F(x_0)$

Distribution function

Properties: **L**$\boldsymbol{et}\ \boldsymbol{X} \sim \boldsymbol{F}(\boldsymbol{x}),\ set\ \boldsymbol{F}(\boldsymbol{x_0} -) = \lim_{\boldsymbol{x} \to \boldsymbol{x_0} -} \boldsymbol{F}(\boldsymbol{x})$.

$if\ a < b$, then:

$$P\{X \le a\} = F(a),\ P\{X > b\} = 1 - F(b),\quad P\{X < a\} = F(a-);$$

$$P\{a < X \le b\} = P\{X \le b\} - P\{X \le a\} = F(b) - F(a),$$

$$P\{X = a\} = F(a) - F(a-);$$

$$P\{a \le X \le b\} = F(b) - F(a-),\ P\{a < X < b\} = F(b-) - F(a),$$

Therefore, as long as the distribution function of random variable $\boldsymbol{X}$ is known, its statistical properties can be fully described.

Examples

Example 2.1.2: Let the distribution function of random variable X in the real domain be

$$F(x)=A+B\arctan x,-\infty<x<+\infty.$$

Calculate: (1) A and B; (2) $P(0<X\leq 1)$.

Answer:(1)

$$\begin{cases} F(-\infty)=0 \\ F(+\infty)=1 \end{cases} \Rightarrow \begin{cases} A+B(-\pi/2)=0, \\ A+B(\pi/2)=1. \end{cases} \Rightarrow \begin{cases} A=1/2 \\ B=1/\pi \end{cases}$$

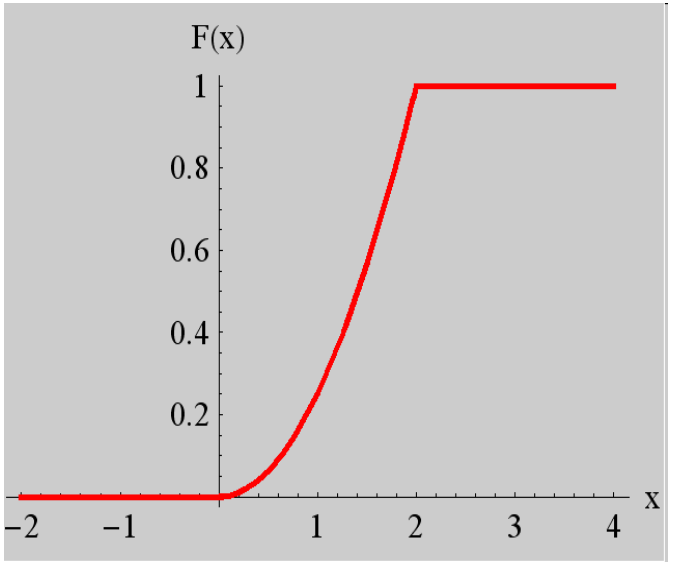
(2) $P(0<X\leq 1)=F(1)-F(0)=1/4.$

Examples

Example 2.1.3: A target is a disk with a radius of 2 meters. Suppose that the probability of shooting any point on a concentric ring is proportional to the area of the ring, and suppose that all shots can hit the target. Let X be the distance between the hitting point and the center of the circle, try to find out the distribution function of random variable X.

Answer:

$$F(x)=\begin{cases}0, & x<0,\\ \dfrac{x^2}{4}, & 0\le x<2,\\ 1, & x\ge 2.\end{cases}$$



Section 2.2 Discrete random variable and its distribution law

- Distribution law of discrete random variable
- Some common distribution function of discrete random variable
- Examples

Discrete random variable

(Definition1.2.1) Let X be an r.v. on the sample space S. If possible values in its range are finite or infinite countable, r.v. X is called a discrete random variable.

Let X be an r.v. whose possible values are x_k, k = 1, 2, 3, · · · : Now we consider the probabilities

$$P\{X = x_k\} = p_k,\ k = 1, 2, \cdots$$

Distribution law of discrete random variable

Its easy to show that the probabilities p_k, $k = 1,2,,$ satisfy the following two properties:

$$(1) p_k \geq 0, k = 1, 2, \cdots$$

$$(2) \quad \sum_{k=1}^{\infty} p_k = 1$$

These two properties are used to judge whether a function is a probability function

- These two properties need to be verified when calculating the distribution law.
- If it is true, it is called the distribution law, otherwise it cannot be said that what is sought is the distribution law.
- Only discrete random variables have distribution law.

Discrete random variable

For convenience, we put X, x_k, p_k into the following table:

X	x_1	x_2	$\dots$	x_n	$\dots$
P	p_1	p_2	$\dots$	p_n	$\dots$

(**Distribution law**) The vector (X, x_k, p_k) or above table is called the **probability mass function**(pmf) or **probability function (p.f)** or **distribution law of *X***.

The probability mass function provides information about the values the r.v. X can take on and the probability of each value! The probability distribution of X shows the value of r.v. X and its distribution completely.

Distribution law of discrete random variable

The distribution law of discrete random variable X can also be marked by

$$X \sim \begin{pmatrix} x_1 & x_2 & \cdots & x_n & \cdots \\ p_1 & p_2 & \cdots & p_n & \cdots \end{pmatrix}$$

X	x_1	x_2	$\cdots$	$x_n \cdots$
p_k	p_1	p_2	$\cdots$	$p_n \cdots$

Distribution law of discrete random variable

Example2.2.1: Out of the 15 products, 2 are defective, and 3 are randomly selected. X represents the number of defective products in 3.. Calculate the distribution law of X.

Answer: X takes value in 0, 1, 2, and:

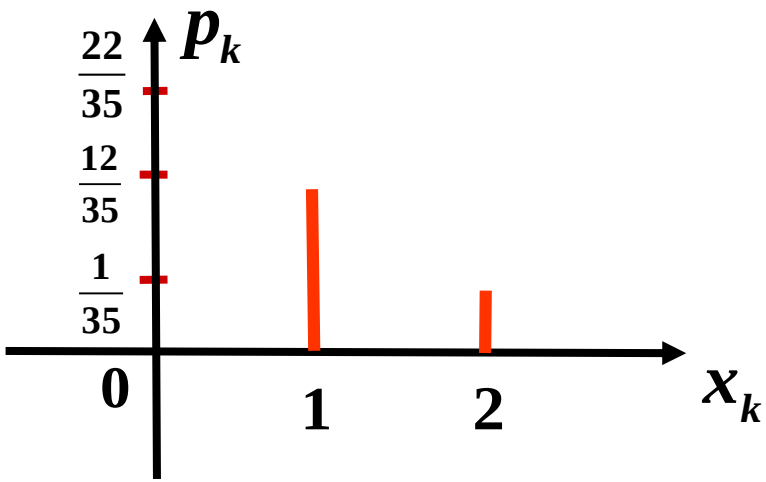
$$P(X=0)=\frac{C_{13}^{3}C_{2}^{0}}{C_{15}^{3}}=\frac{22}{35}$$

$$P(X=1)=\frac{C_{13}^{2}C_{2}^{1}}{C_{15}^{3}}=\frac{12}{35}$$

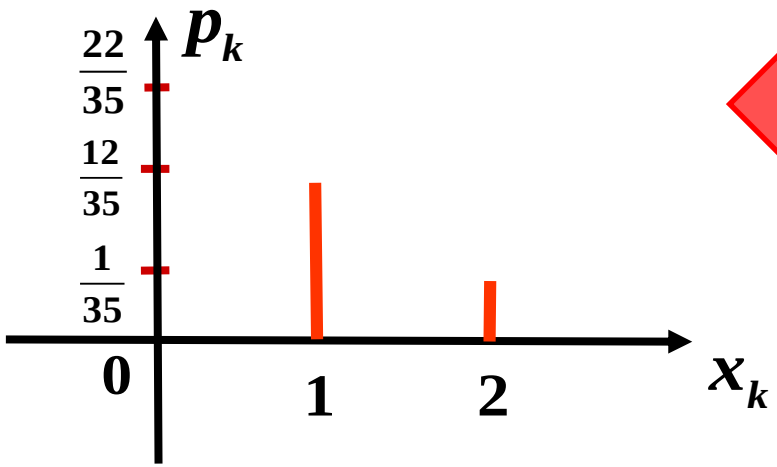
$$P(X=2)=\frac{C_{13}^{1}C_{2}^{2}}{C_{15}^{3}}=\frac{1}{35}$$

(obviously, $P_k>0$, $\sum_{k=0}^{3}P_k=1$)

X	0	1	2
P_k	$\frac{22}{35}$	$\frac{12}{35}$	$\frac{1}{35}$

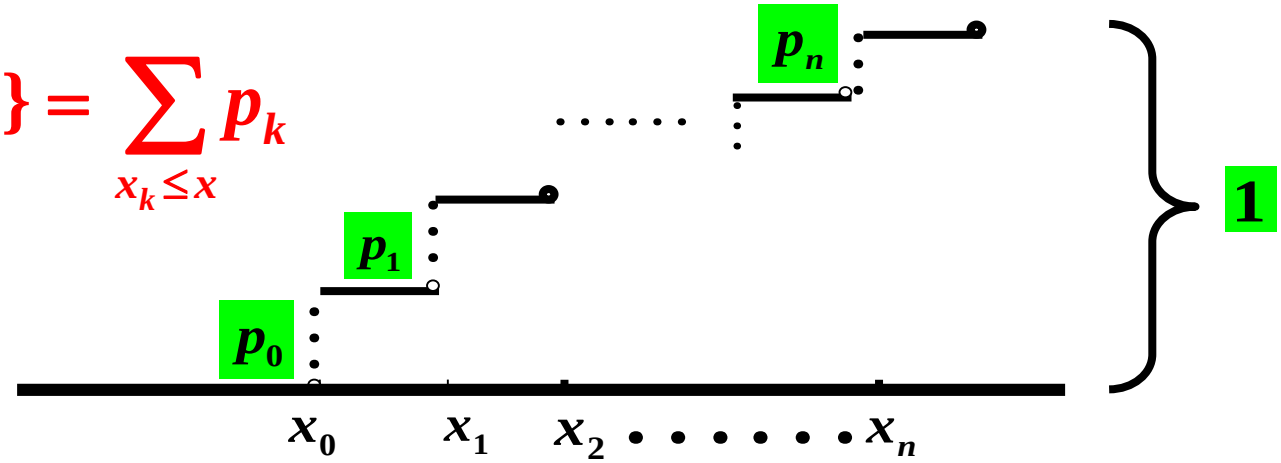


Distribution law



$$p_k = P\{X = x_k\}$$

$$F(x) = P\{X \le x\} = \sum_{x_k \le x} p_k$$



Distribution Function

The distribution function of discrete random variable:

$$F(x) = P\{X \le x\} = \sum_{x_k \le x} p_k = \sum_{x_k \le x} P(X = x_k).$$

The relationship between the distribution law and function of discrete random variable:

Distribution Law $p_k = P\{X = x_k\}$

$$\Updownarrow$$

(Cumulative)
Distribution Function $F(x) = P\{X \le x\} = \sum_{x_k \le x} p_k$

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Uniform Distribution

If the probability distribution of $r.v.\ X$ is

X	a_1	$a_2 \cdots a_n$
p_k	$\frac{1}{n}$	$\frac{1}{n} \cdots \frac{1}{n}$

in which $(a_i \neq a_j),(i \neq j)$, then X is a **Uniform distribution**.

Example: tossing a dice, X is the number on the top.

X	1	2	3	4	5	6
p_k	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

(0-1) Distribution

If possible values of $r.v.$ X are **only 0 and 1**, and its probability distribution is

X	0	1
p_k	$1-p$	p

Then X obeys (0-1) distribution or a two-point distribution, marked by $X \sim b(1, p)$

(0-1) Distribution

Example: Tossing a coin, observing the frontage or the reverse side on the top.

NOTE:
A two-point distribution is the simplest distribution. Any random phenomenon with only two possible outcomes, such as whether the baby will be a boy or a girl, whether it will rain tomorrow, whether a seed will germinate, etc., is a two-point distribution.

Binomial Distribution

Example2.2.2: If the probability of giving birth to a boy is p , and the probability of giving birth to a girl is q=1-p, let X represent the number of "boys" among the four babies born at random.

Calculate: the probability distribution of X.

X takes value in： 0, 1, 2, 3, 4.

$$P\{X=k\}=C_4^k\, p^k (1-p)^{4-k}, \quad k=0,1,2,3,4$$

Binomial Distribution

Example 2.2.3: Throw an even die three times, let X indicates the number of times that "4" appears in 3 occurrences. Calculate the probability distribution law of X.

X take values in： 0, 1, 2, 3.

$$P\{X=k\}=C_3^k\left(\frac{1}{6}\right)^k\left(\frac{5}{6}\right)^{3-k},\quad k=0,1,2,3$$

Binomial Distribution

According to the addition principle, the probability that event ***X*** happens *k* times in the *n*-th Bernoulli test is:

$$\underbrace{p\cdot p\cdots\cdots p}_{k}\cdot\underbrace{(1-p)(1-p)\cdots(1-p)}_{n-k}=p^k(1-p)^{n-k}$$

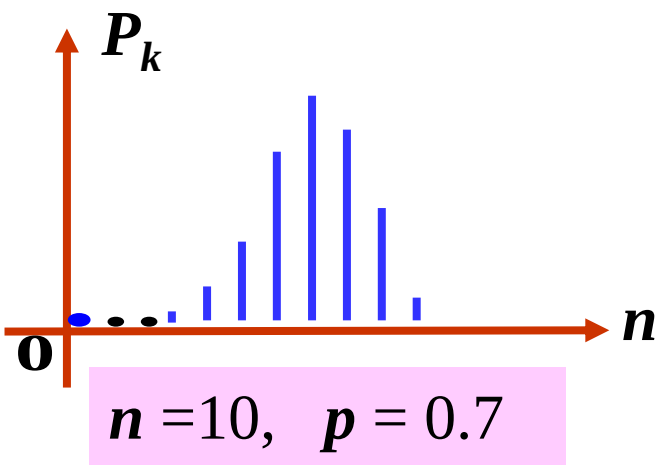
$$P_n(k)=C_n^k p^k(1-p)^k \qquad k=0,1,2\cdots n$$

$$P(X=k)\geq 0, \qquad \sum_{k=0}^{n}C_n^k p^k(1-p)^{n-k}=(p+q)^n=1$$

Probability *Pn (k)* is equal to the coefficient *of* x^k in the expansion of binomial **[(1-*p*) + *px*] ^{n}**, which is also the origin of the name of binomial distribution.

Binomial Distribution

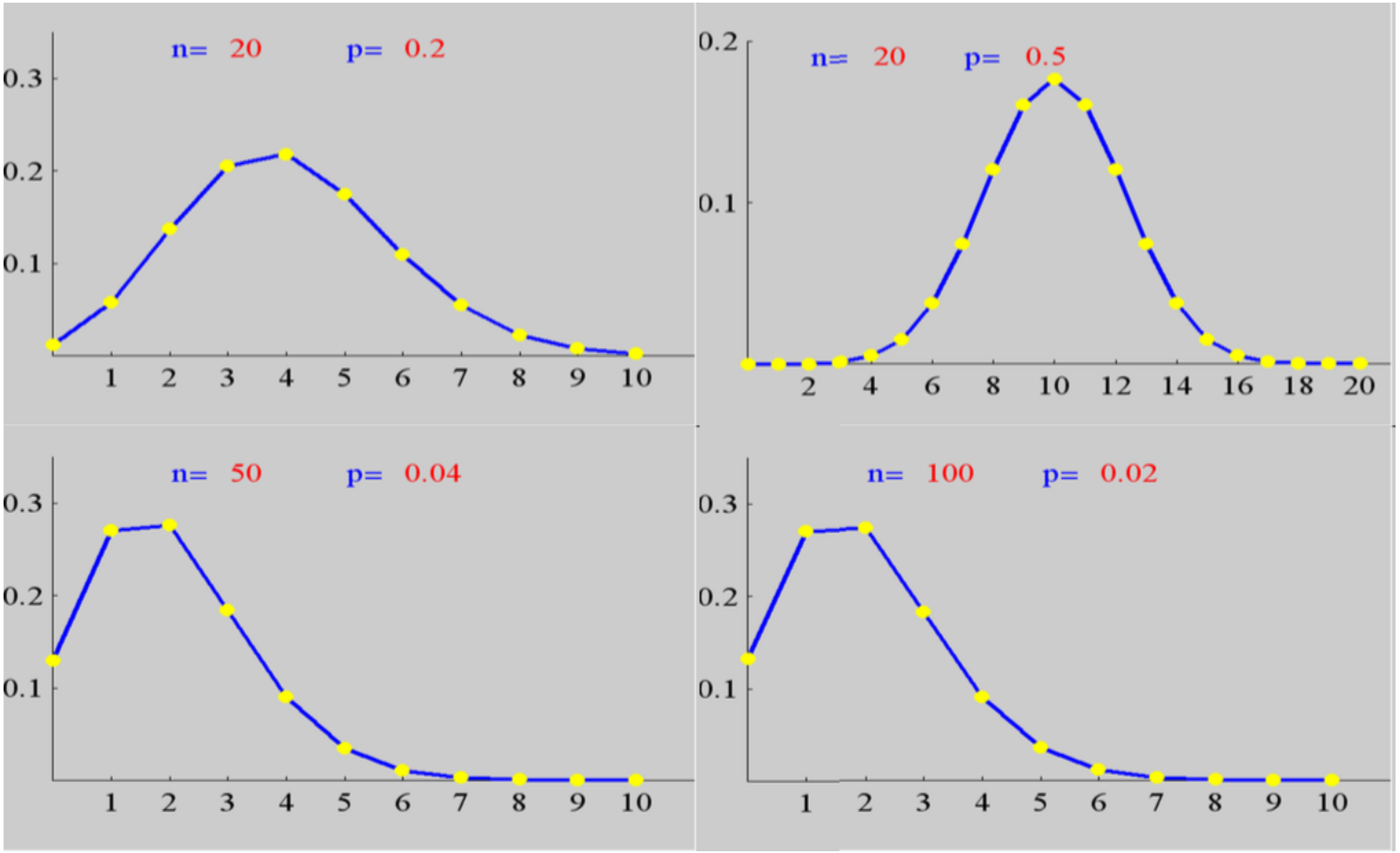
If $r.v.$ X obeys $P\{X=k\}=C_n^k p^k q^{n-k}, k=0,1,2,\ldots n$ in which q=1-p, then $\boldsymbol{X}$ **obeys Binomial distribution with *n* and *p*** as parameters, marked by $\boldsymbol{X \sim B(n,p)}$.



For fixed ***n*** and ***p***, when k increases, the probability ***P(x=k)*** first increases until it reaches the maximum, and then decreases monotonically

Especially, n=1, $X \sim B(n,p) \rightarrow X \sim b(1,p)$

Binomial Distribution



Binomial Distribution

Example 2.2.3: Among the 100 products, there are 5 defective products, which are now taken back 3 times with put it back, one at a time.

Calculate: the probability of exactly two defective products among the three taken products.

Answer:

The probability of getting defective products in each test is 0.05.

Let X: the number of defective products in the three selected,

then, $X \sim B(3, 0.05)$

$$P(X=2) = C_3^2 (0.05)^2 (0.95) = 0.007125$$

Binomial Distribution

Example2.2.4: A novice is practicing shooting, assuming that the probability of hitting the target in one shot is 0.02, this person is practicing 400 shots. Calculate the probability of at least two shots hitting the target.

$$P\{X \geq 2\} = 1 - P\{X = 0\} - P\{X = 1\}$$

$$= 1 - (0.98)^{400} - 400(0.02)(0.98)^{399}$$

$$= 0.9972.$$

Binomial Distribution

Example2.2.5: If the probability of death of each person participating in life insurance in a year is 0.005, there are 10000 such people participating in life insurance.

Calculate: among these insurers in the coming year:
(1) The probability of 10 deaths
(2) The probability that the death toll will not exceed 10.

Answer: Let X: the number of deaths among these insurers in the next year， then：

$$X \sim B(10000,\ 0.005)$$

(1) The probability of 10 deaths is:

$$P(X=10)=C_{10000}^{10}(0.005)^{10}(0.995)^{9990}$$

(2) The probability that the number of deaths does not exceed 10 is:

$$P(X\le 10)=\sum_{k=0}^{10}C_{10000}^{k}(0.005)^{k}(0.995)^{10000-k}$$

These calculations are so very troublesome. An approximate formula is given when n is large and p is small, that is, the **Possion approximation of binomial distribution.**

Poisson's theorem

Suppose $X \sim B(n, p), \lambda > 0$ is a constant, and $\lambda = nP_n$, Then, for any fixed nonnegative integer k,

$$\lim_{n\to\infty} P_n(k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Binomial Distribution Examples

Generally, $\dfrac{\lambda^k e^{-\lambda}}{k!}$ was usually used to approximate $P_n(k)$.

And, when $n \geq 20, \quad p \leq 0.05$, the approximate is good;

When $n \geq 100, \ np \leq 10$, the approximate will be excellent.

Binomial Distribution Examples

Example2.2.5: If the probability of death of each person participating in life insurance in a year is 0.005, there are 10000 such people participating in life insurance.

Calculate: among these insurers in the coming year:
(1) The probability of 10 deaths
(2) The probability that the death toll will not exceed 10.

$$\because \quad \lambda = 10000 \times 0.005 = 50$$

$$(1) \therefore \quad P(X = 10) \approx \frac{(50)^{10} \cdot e^{-50}}{10!}$$

$$(2) \quad P(X < 10) \approx \sum_{k=0}^{10} \frac{(50)^k e^{-50}}{k!}$$

Binomial Distribution Examples

If "the probability of death per person "was changed from 0.005 to 0.0005, then $\lambda = 5$,

Thus, $$P(X=10)=\frac{5^{10}\cdot e^{-5}}{10!}=\frac{9765625\times 0.0067379}{3628800}$$

$$=0.01813$$

$$P(X<10)=1-P(X\geq 10)\approx 1-\sum_{k=10}^{\infty}\frac{5^{10}e^{-5}}{10!}$$

$$=1-0.031828$$

$$=0.96817$$

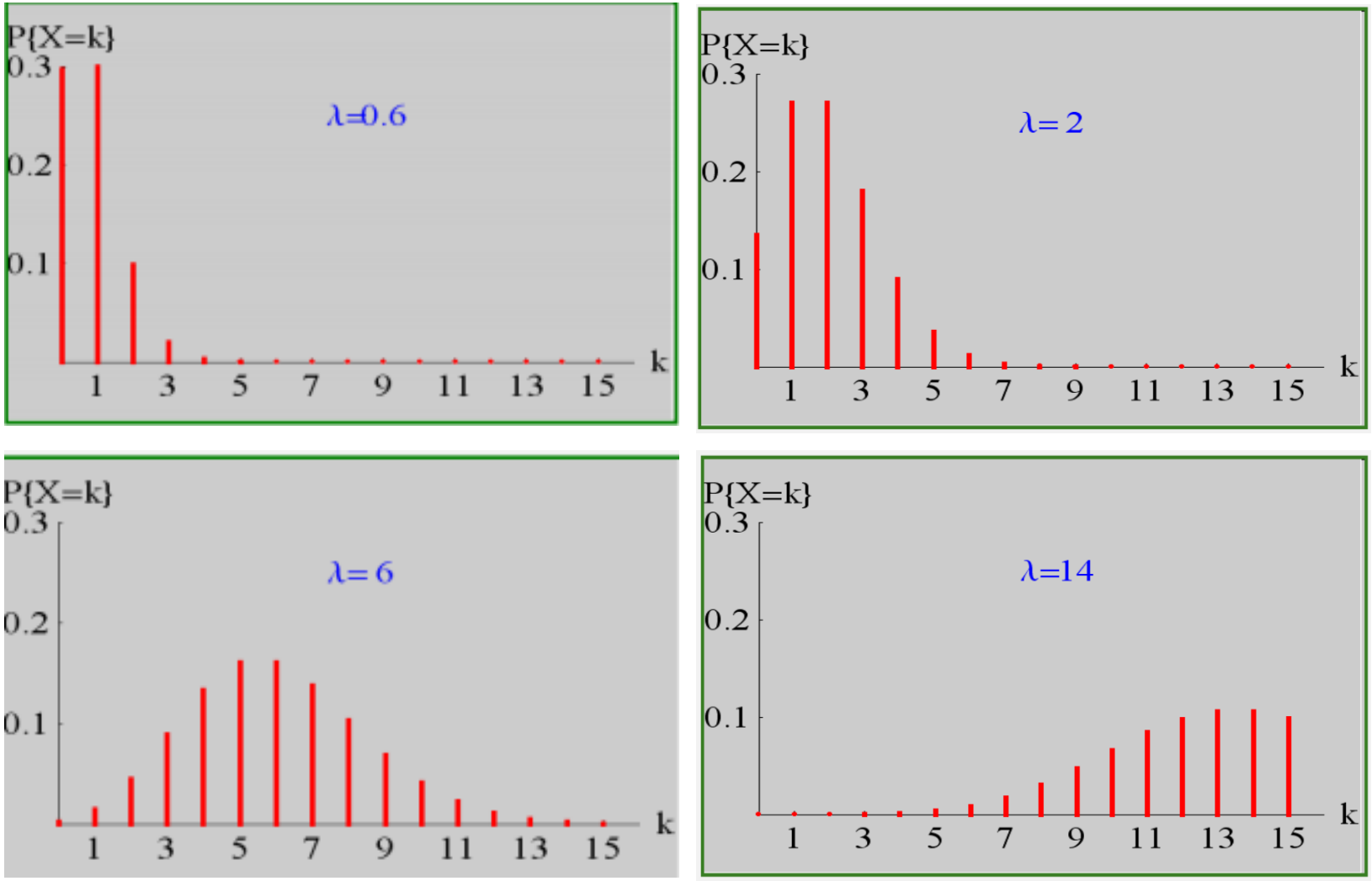
Poisson Distribution

Poisson Distribution: Suppose $\boldsymbol{X}$ be a discrete r.v., if the probability distribution of $\boldsymbol{X}$ is

$$P\{X = k\} = \frac{\lambda^k e^{-\lambda}}{k!}, k = 0,1,2 \ldots$$

in which $\boldsymbol{\lambda > 0}$, then $\boldsymbol{X}$ is obey a Poisson distribution with parameter $\boldsymbol{\lambda}$, and we denote $\boldsymbol{X \sim P(\lambda)}$.

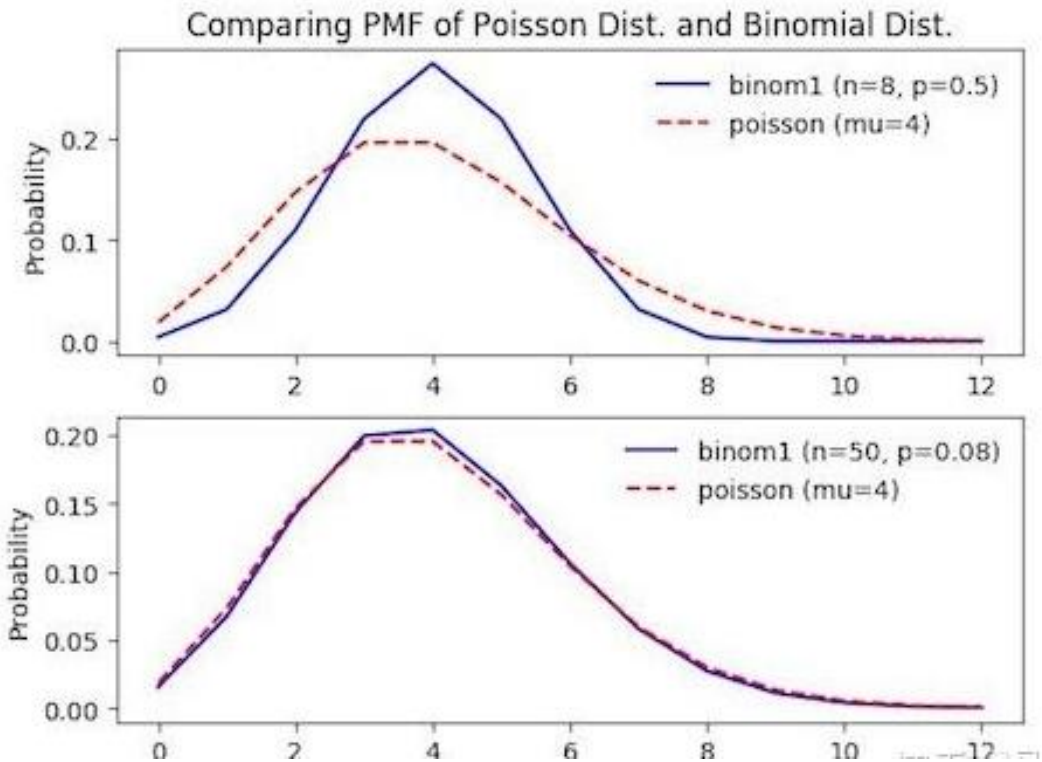
Poisson Distribution



Poisson Distribution

Relationships between **Poisson & Binomial distribution :**

For a Binomial distribution when $\boldsymbol{n} \rightarrow \infty$, $np=\boldsymbol{\lambda}$, then



$$\lim_{n\to\infty} P\{X=k\} = \frac{\lambda^k}{k!} e^{-\lambda}$$

Poisson Distribution

The background and application of Poisson Distribution:

In the early 1900s, two scientists, Rotherford and Gekko, were observing and analyzing the number of particles emitted by radioactive materials, They conducted 2,608 observations (7.5 seconds each) and found that the number X of particle emitted from a radioactive substance during a specified period of time follows a Poisson distribution.

Poisson Distribution

The background and application of Poisson Distribution:

In biology, medicine, industrial statistics, insurance science, Poisson distribution is common. Earthquakes, volcanic eruptions, catastrophic floods all obey the Poisson distribution.

What is the Criterion of Poisson Distribution?

Poisson Distribution

Example 2.2.6: There is a busy bus station where a large number of cars pass every day. Let the probability of each car having an accident at a certain time of day is 0.0001, 1000 cars pass during one hour. So, what's the probability that at least 2 accidents occur in one hour?

Solution:

Let $X = \{x \text{ accidents occur in one hour}\}$, then we have $\boldsymbol{X \sim B(1000, 0.0001)}$.

Poisson Distribution

Solution:

$$X \sim B(1000, 0.0001)$$

Applying Binomial distribution:

$$P\{X \geq 2\} = 1 - P\{X = 0\} - P\{X = 1\}$$

$$= 1 - 0.9999^{1000} - \binom{1000}{1} \cdot 0.0001 \cdot 0.9999^{999}$$

Applying Poisson theorem:

$$\lambda = 1000 \times 0.0001 = 0.1,$$

$$P\{X \geq 2\} \approx 1 - \frac{e^{-0.1}}{0!} - \frac{0.1 \cdot e^{-0.1}}{1!} = 0.0047.$$

Geometric Distribution

In the Bernoulli experiment sequence, the probability of event A occurring is $\boldsymbol{p}$. The experiment continues **until** the first occurrence of event A. The number of experiments conducted until the occurrence of event A is denoted as $\boldsymbol{X}$.

Obviously, the value range of $\boldsymbol{X}$ is all positive integers, determined by **independence**,

$$P\{X = k\} = q^{k-1}p, \qquad q = 1 - p, k = 1, 2 \ldots$$

Since $\{q \wedge (k-1)\ p\}$ is a geometric sequence, the random variable $\boldsymbol{X}$ with the above probability distribution is said to follow a geometric distribution with a parameter $\boldsymbol{p}$, abbreviated as: $X \sim G\ (p)$

Geometric Distribution

Example: Set $\boldsymbol{X \sim G(p)}$, Prove： for any interger m, n,

$$\boldsymbol{P(X > m + n | X > m) = P(X > n)}$$

Prove:

$$P(X > m) = \sum_{k=m+1}^{\infty} q^{k-1} p = \frac{q^m p}{1-q} = q^m$$

$$P(X > m + n | X > m) = \frac{P(\{X > m + n\} \cap \{X > m\})}{P(X > m)}$$

$$= \frac{P(X > m + n)}{P(X > m)}$$

$$= \frac{q^{m+n}}{q^m} = q^n = P(X > n)$$

Memoryless Property

Hypergeometric Distribution

This distribution occurs quite often and is suitable in describing situations of the following type: $\boldsymbol{N}$ white balls are thoroughly mixed with $\boldsymbol{M}$ black balls identical. From these M+$\boldsymbol{N}$ objects, $\boldsymbol{n}$ balls are drawn without replacement. Let $\boldsymbol{X}$ be the number of white balls.

Then the r.v. $\boldsymbol{X}$ takes on the values $0, 1, 2, \cdots; \min\{N, n\}$

$$\boldsymbol{P\{X = k\} = \frac{C_N^k C_M^{n-k}}{C_{N+M}^n}, \qquad 0 \le k \le min(n, N)}$$

Hypergeometric Distribution

If we take $\boldsymbol{n}$ balls with replacement, it's a Binomial Distribution,

$$P\{X=k\}=C_n^k\left(\frac{N}{N+M}\right)^k\left(\frac{M}{N+M}\right)^{n-k}, \qquad 0\le k\le n$$

That is, when $\mathbf{N}\to\infty$, $M\to\infty$, $\frac{N}{N+M}\to p$

$$P\{X=k\}=\frac{C_N^k C_M^{n-k}}{C_{N+M}^n}\approx C_n^k\left(\frac{N}{N+M}\right)^k\left(\frac{M}{N+M}\right)^{n-k}$$

$$\frac{C_N^k C_M^{n-k}}{C_{N+M}^n}\to C_n^k p^k(1-p)^{n-k}$$

Brief summary

$$X \sim b(1,\ p)$$

$$\uparrow \quad n=1$$

$$P\{X=k\}=\frac{C_N^k C_M^{n-k}}{C_{N+M}^n} \xrightarrow{\substack{N,\ M\uparrow\uparrow \\ N\downarrow\downarrow}} C_n^k p^k(1-p)^{n-k} \qquad X \sim b(n,\ p)$$

$$\downarrow \quad \begin{array}{c} n\uparrow\uparrow,\ p\downarrow\downarrow \\ np=\lambda \end{array}$$

$$P\{X=k\}=\frac{\lambda^k}{k!}e^{-\lambda}, k=0,1,\ldots.$$